

Density-Invariant Difference-DMD on Waddington-OT

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1. Dynamical Model

I used Waddington-OT’s mean trajectories as data. Since the cell expressions were under the influence of OKSM, we can model it as:

$$x_{k+1} = (I + A)x_k + b_k \quad (1)$$

where $x_k \in \mathbb{R}^G$ is a mean centred trajectory, A is the intrinsic unforced gene regulatory network operator, and b_k is the external control input (forcing) applied to the system at time step k .

2. Difference-DMD for State Isolation

The model assumes that external forcing b_k is uniformly applied to all parallel lineages at a given time point, so we can isolate A:

$$x'_{k+1} - x_{k+1} = (I + A)(x'_k - x_k) \quad (2)$$

where x' and x are two parallel trajectories. Applying half-rank truncated Dynamic Mode Decomposition to these trajectory pair differences obtain a set of models A_l .

3. Geometric Fusion

We then obtain the global model via fusing these matrices:

$$A_{fused} = \left(\sum_l w_l A_l \right) \left(\sum_l w_l P_l + \epsilon I \right)^{-1} \quad (3)$$

This fused matrix is the optimal solution to:

$$\min_A \sum_l w_l \| (A - A_l) P_l \|_F^2 \quad (4)$$

where P_l is the orthogonal projector onto the lineage’s subspace, and $w_l = 1 / \sum_j \text{tr}(P_l P_j)$ acts as an inverse-density penalty to prevent redundant failure states (e.g., stagnant stroma) from mathematically diluting rare successful trajectories. The fused model has $r^2 = 0.87$ for individual trajectory difference series.

4. Validation

Beyond the predictiveness on lineage differences, by obtaining A we can also recover the forcing term b_k :

$$b_k = x_{k+1} - (I + A_{fused})x_k \quad (5)$$

The variance of the recovered b_k from data is minimal compared to overall state transitions, so at least compared to states, the control is relatively per lineage constant. In linear control theory, $b_k = Bu_k$, where u_k are Transcription Factors and the matrix B translates TF binding into genome-wide expression influences. From this model, one expect the variance of b_k over time to be roughly 4 dimensional or less, which is verified:

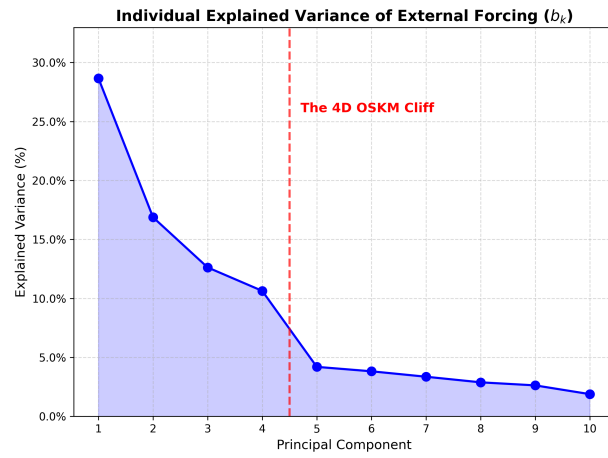


Figure 1: Scree Plot